



Problem solving questions for A Level Physics

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Writing problem solving questions



- **Audience:** A Level students who enjoy Physics & Maths and want to apply their knowledge from the course to challenging problems
- **Aims:**
 - To write problems which test a student's problem-solving skills
 - To reinforce the connection between Physics and Maths
 - To improve fluency in applying and manipulating equations
 - To consolidate A Level concepts and using them in unfamiliar situations

What makes a good question?

- Something a bit different:
 - Unusual setups / unfamiliar applications
- Multiple-step reasoning (may be more than one valid method):
 - 'first use this equation, then this assumption, then another equation'
- Provides enough information to clearly define the setup, but also concise
- Neat answer (not always necessary)

Step 1: Topic and idea

Topic:

- Do they know the content/have the mathematical skills? Should be topics covered on the specification – no extra reading
- Choose a pair of topics/concepts, and think of ways to combine them

Idea:

- Often the most difficult part!
- Choose an interesting set-up/geometry
- Look at past papers/Olympiad papers
- Random inspiration (write down the idea for later)

Step 2: Do the algebra



- The question may fail:
 - Not the right level
 - Equations too complicated
 - Too many assumptions
 - Back to Step 1!
- Decide whether the answer should be numeric or symbolic
- How much algebra? I usually aim for $\frac{1}{2}$ A4 page of algebra for a typical challenge question

Step 3: Write it up

- Decide whether you will give the diagram in the question, or expect the student to draw their own based on the information given
- State assumptions, variables, constants
- How long?
 - Typically 1–3 parts; too much structure will guide the student along a particular method
 - Decide how long you want a student to spend on your question, one minute/one hour?
- ‘Show that’ question?

Step 4: Review

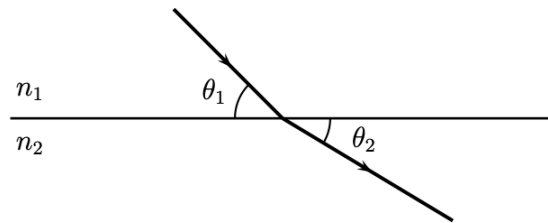


- Give to a student or another teacher (or AI?)
- Question might need rewording / diagram might need changes etc.
- Useful to see another person's method
- If numeric, check if values given are sensible
- Do you need to include a hint?

Examples from BPhO Round 0 2025

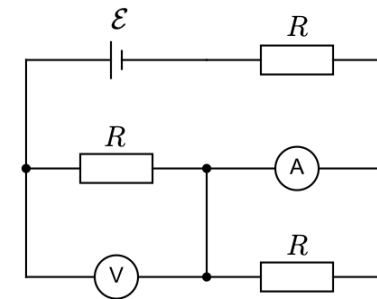


2. The diagram below shows the refraction of a light ray travelling from a medium with refractive index n_1 into a medium with refractive index n_2 . Which equation relates the angles θ_1 and θ_2 ?



- A** $n_1 \sin \theta_1 = n_2 \sin \theta_2$ **B** $n_1 \sin \theta_2 = n_2 \sin \theta_1$ **C** $n_1 \cos \theta_1 = n_2 \cos \theta_2$ **D** $n_1 \cos \theta_2 = n_2 \cos \theta_1$ **E** $n_1 \sin \theta_1 = n_2 \cos \theta_2$

8. A cell of emf $\mathcal{E}$ (and negligible internal resistance), is connected to an ideal voltmeter, an ideal ammeter and three resistors R as shown below. What are the readings on the voltmeter and ammeter respectively?



- A** $\frac{\mathcal{E}}{3}, \frac{\mathcal{E}}{3R}$ **B** $\frac{\mathcal{E}}{3}, \frac{\mathcal{E}}{2R}$ **C** $\frac{\mathcal{E}}{3}, \frac{\mathcal{E}}{R}$ **D** $\frac{\mathcal{E}}{2}, \frac{\mathcal{E}}{3R}$ **E** $\frac{\mathcal{E}}{2}, \frac{\mathcal{E}}{2R}$

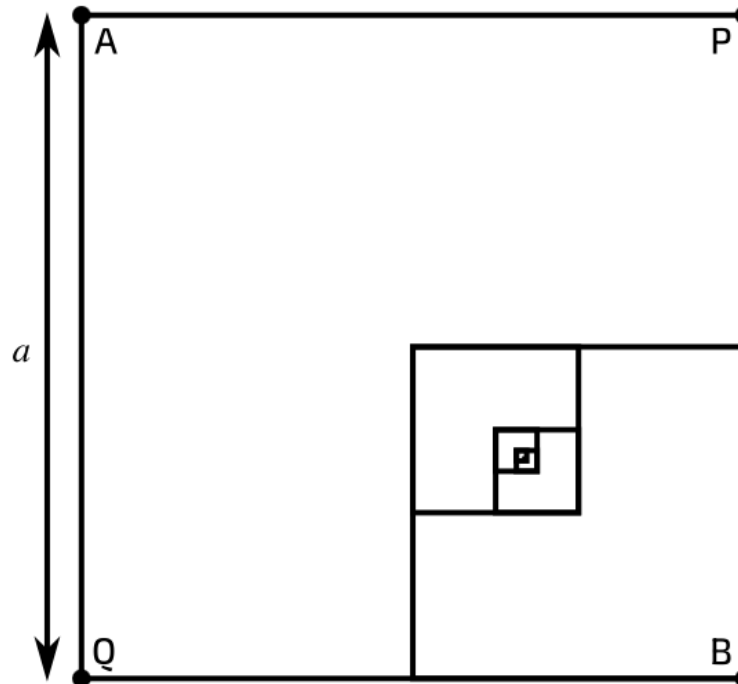
Example: Resistor Recursion (A Level C2)



The pattern in **Figure 1** shows a square of side length a .

From the midpoints of two adjacent sides, straight lines of equal length are drawn to divide the square into the ratio 3 : 1 by area. The process is repeated infinitely within the smaller piece of the square each time.

This pattern is recreated using a thin wire, with uniform resistance per unit length k . Although it is not physically possible to perfectly recreate this infinitely repeating pattern, we are interested in the limit of large numbers of repetitions.



Projectile on a String (A Level C3)



Projectile on a String



Subject & topics

Physics | Mechanics | Dynamics

Status

Correct

Stage & difficulty

A Level Challenge 3

University Practice 1



A particle of mass m is attached to one end of a light inextensible string of length l ; the other end is fixed at a point O. The particle is initially at point P, which is a horizontal distance l from O.

The particle is projected vertically upwards at speed v , and in its subsequent motion, it passes through O.

Find an expression for v in terms of m , g and l .

Example: Moving Fringes (A Level C3)

A double slit aperture is printed on a thin, transparent and flexible film. The film is stretched from the sides with a time-varying force so that the slit separation d is given as a function of time t as

$$d(t) = d_0 \left(1 + \frac{2}{7} \sin \omega t \right),$$

where d_0 and ω are constants.

Coherent monochromatic light of wavelength λ passing normally through the aperture produces a diffraction pattern on a screen placed a distance L away from it. The fringes can be assumed to be evenly spaced on the screen.

Find an expression for the maximum speed of the 3rd order maxima as viewed on the screen.